Case Study 4: Attendance Recovery Computation

1. Problem Statement

To develop a Python program that calculates a student's current attendance percentage and determines the minimum number of consecutive classes the student must attend to reach the required attendance eligibility percentage.

2. Objectives

- To calculate the current attendance percentage.
- To check whether a student meets the required attendance percentage.
- To calculate the minimum number of classes needed to reach the eligibility target.
- To understand the use of Python loops and conditional statements.

3. Assumptions

For this assignment, the following sample values and rules are used.

Component	Value
Total classes conducted	Entered by the user
Classes attended	Entered by the user
Required attendance	75%

Future attendance scenario	Student attends every upcoming class
Valid attendance	Classes attended cannot exceed total classes

4. Algorithm

1. Start the program.
2. Input the total classes conducted and classes attended.
3. Validate the input values.
4. Calculate the current attendance percentage.
5. Check whether attendance is at least 75%.
6. If attendance is below 75%, calculate the minimum additional classes needed.
7. Display the current attendance and required classes.
8. Stop the program.

5. Flowchart

Start
Input total classes and attended classes
Calculate attendance percentage
Is attendance at least 75%?
Yes
Display eligible
No
Calculate required classes
Stop

6. Pseudocode

```
None
START

INPUT total_classes
INPUT attended_classes
```

```
IF total_classes <= 0 OR attended_classes < 0
   OR attended_classes > total_classes:
    DISPLAY "Invalid input"
ELSE:
    attendance = attended_classes / total_classes * 100

    DISPLAY attendance

    IF attendance >= 75:
        DISPLAY "Attendance eligibility achieved"
    ELSE:
        future_classes = 0

        WHILE (attended_classes + future_classes)
              / (total_classes + future_classes) < 0.75:
            future_classes = future_classes + 1

        DISPLAY future_classes

STOP
```

7. Complete Python source code

Save the file as `attendance_recovery.py`.

None

```
# Attendance Recovery Computation

REQUIRED_ATTENDANCE = 75

total_classes = int(input("Enter total classes conducted: "))
attended_classes = int(input("Enter classes attended: "))

# Validate input
```

```
if total_classes <= 0:
    print("Invalid input! Total classes must be positive.")

elif attended_classes < 0 or attended_classes > total_classes:
    print("Invalid input! Check the number of classes attended.")

else:
    # Calculate current attendance
    attendance = (attended_classes / total_classes) * 100

    print("\nCurrent Attendance:",
          round(attendance, 2), "%")

    if attendance >= REQUIRED_ATTENDANCE:
        print("Status: Attendance eligibility achieved.")
        print("Additional classes required: 0")

    else:
        future_classes = 0

        # Assume the student attends every future class
        while (
            (attended_classes + future_classes)
            / (total_classes + future_classes)
        ) * 100 < REQUIRED_ATTENDANCE:
            future_classes += 1

        print("Status: Attendance below 75%.")
        print("Minimum consecutive classes to attend:",
              future_classes)

        final_attendance = (
            (attended_classes + future_classes)
            / (total_classes + future_classes)
        ) * 100
```

```
print("Attendance after attending these classes:",
      round(final_attendance, 2), "%")
```

8. Test cases and expected output

Test case	Total classes	Attended	Current attendance	Additional classes
1	100	60	60%	60
2	100	70	70%	20
3	100	80	80%	0
4	50	40	80%	0

Sample execution

Input:

```
None
Enter total classes conducted: 100
Enter classes attended: 60
```

Output:

```
None
Current Attendance: 60.0 %
Status: Attendance below 75%.
```

```
Minimum consecutive classes to attend: 60
Attendance after attending these classes: 75.0 %
```

Explanation: The student attended 60 out of 100 classes. If the student attends the next 60 classes without missing any, attendance becomes \(120/160 \times 100 = 75\%\).

9. Output screenshot

```
Enter "help" below or click "Help" above for more information.
>>
========== RESTART: C:/Users/YOGITHA/OneDrive/Desktop/4.Attendence.py ==
Enter total classes conducted: 100
Enter classes attended: 60

Current Attendance: 60.0 %
Status: Attendance below 75%.
Minimum consecutive classes to attend: 60
Attendance after attending these classes: 75.0 %
>>
============== RESTART: C:/Users/YOGITHA/OneDrive/Desktop/4.Attendence.p
Enter total classes conducted: 100
Enter classes attended: 70

Current Attendance: 70.0 %
Status: Attendance below 75%.
Minimum consecutive classes to attend: 20
Attendance after attending these classes: 75.0 %
>>
```

```python
# Attendance Recovery Computation

REQUIRED_ATTENDANCE = 75

total_classes = int(input("Enter total classes conducted: "))
attended_classes = int(input("Enter classes attended: "))

# Validate input
if total_classes <= 0:
    print("Invalid input! Total classes must be positive.")

elif attended_classes < 0 or attended_classes > total_classes:
    print("Invalid input! Check the number of classes attended.")

else:
    # Calculate current attendance
    attendance = (attended_classes / total_classes) * 100

    print("\nCurrent Attendance:",
          round(attendance, 2), "%")

    if attendance >= REQUIRED_ATTENDANCE:
        print("Status: Attendance eligibility achieved.")
        print("Additional classes required: 0")

    else:
        future_classes = 0

        # Assume the student attends every future class
        while (
            (attended_classes + future_classes)
            / (total_classes + future_classes)
        ) * 100 < REQUIRED_ATTENDANCE:
            future_classes += 1

        print("Status: Attendance below 75%.")
        print("Minimum consecutive classes to attend:",
              future_classes)

        final_attendance = (
            (attended_classes + future_classes)
            / (total_classes + future_classes)
        ) * 100

        print("Attendance after attending these classes:",
              round(final_attendance, 2), "%")
```

10. Conclusion

The Attendance Recovery Computation program calculates current attendance and determines the minimum number of consecutive classes required to reach the 75% eligibility target. It demonstrates Python variables, arithmetic operators, conditional statements, and `while` loops.

Case Study 4 is ready! You can copy these sections into your assignment report. Next is Case Study 5: ATM Transaction Sequence Simulation.